

SECTION B: CIRCULAR MEASURE AND TRIGONOMETRIC FUNCTIONS

1. Evaluate $\frac{\sin^2 \theta}{\cos^2 \theta - 1}$.

A. -1

B. 1

C. 2

D. -2

2. Simplify $\frac{\sin^2 x}{\tan x}$.

A. $\sin x$

B. $\cos x$

C. $\sin x \cos x$

D. $\sin^2 x \cos^2 x$

3. If $\tan \frac{1}{2}x = b$, express $\tan x$ in terms of b .

A. $2b$

B. $1 - b^2$

C. $\frac{1-b^2}{2b}$

D. $\frac{2b}{1-b^2}$

4. If $\tan \frac{1}{2}x = f$, what is $\sin x$ in terms of f ?

A. $\frac{2f}{1+f^2}$

B. $\frac{2f}{1+f}$

C. $\frac{2f^2}{1+f}$

D. $\frac{2f^2}{1+f^2}$

5. If $\tan \frac{1}{2}x = k$, what is $\cos x$ in terms of k ?

A. $\frac{2k^2}{1+k^2}$

- B. $\frac{1-2k^2}{1+k^2}$
 C. $\frac{1-k^2}{1+k^2}$
 D. $\frac{1+k^2}{1-k^2}$

6. What is the result of $1 + \tan^2 \theta$?

- A. $\sin^2 \theta$
 B. $\sec^2 \theta$
 C. $\cos^2 \theta$
 D. $\tan^2 \theta$

7. Evaluate $1 + \cot^2 \theta$.

- A. $\sin^2 \theta$
 B. $\operatorname{cosec}^2 \theta$
 C. $\cos^2 \theta$
 D. $\tan^2 \theta$

8. Simplify $\sin^2 p + (1 + \cos^2 p)^2$.

- A. $2(1 + \cos p)$
 B. $2 \cos p$
 C. $1 + \cos p$
 D. $\cos p - 1$

9. Evaluate $\frac{1+\sin \emptyset}{1+\cos \emptyset} \times \frac{1+\sec \emptyset}{1+\operatorname{cosec} \emptyset}$.

- A. $\sin \emptyset$
 B. $\cos \emptyset$
 C. $\tan \emptyset$
 D. $\cot \emptyset$

10. Expand $\tan (45^\circ + A)$.

- A. $\frac{1+\tan A}{1+\tan A}$
 B. $\frac{1-\tan A}{1-\tan A}$

- C. $\frac{1+\tan A}{\tan A}$
 D. $\frac{1+\tan A}{1-\tan A}$

11. Evaluate $\tan (90^\circ + A)$.

- A. 0
 B. ∞
 C. -1
 D. 1

12. If $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$; $\sin 30^\circ = \frac{1}{2}$; $\cos 30^\circ = \frac{\sqrt{3}}{2}$, evaluate $\sin 75^\circ$.

- A. $\frac{\sqrt{2}+\sqrt{6}}{4}$
 B. $\frac{\sqrt{2}-\sqrt{6}}{4}$
 C. $\frac{\sqrt{6}-\sqrt{2}}{4}$
 D. $\frac{\sqrt{2}+\sqrt{3}}{4}$

13. If $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$; $\sin 30^\circ = \frac{1}{2}$; $\cos 30^\circ = \frac{\sqrt{3}}{2}$, evaluate $\cos 75^\circ$. A.

- $\frac{\sqrt{6}+\sqrt{2}}{4}$
 B. $\frac{\sqrt{2}+\sqrt{3}}{4}$
 C. $\frac{\sqrt{6}-\sqrt{2}}{4}$
 D. $\frac{\sqrt{2}-\sqrt{6}}{4}$

14. Find $\cos 105^\circ$, if $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$; $\cos 60^\circ = \frac{1}{2}$

- A. $\frac{\sqrt{2}+\sqrt{6}}{4}$

B. $\frac{\sqrt{2}-\sqrt{6}}{4}$

C. $\frac{\sqrt{2}-\sqrt{6}}{5}$

D. $\frac{\sqrt{2}-\sqrt{3}}{4}$

15. Which of these evaluates to 1?

A. $\sec^2 x - \tan^2 x$

B. $\sec^2 x + \tan^2 x$

C. $\tan^2 x - \sec^2 x$

D. $\tan x - \sec^2 x$

16. Which of these evaluates to 1?

A. $\cot y - \operatorname{cosec} y$

B. $\cot^2 y - \operatorname{cosec}^2 y$

C. $\operatorname{cosec} y - \cot y$

D. $\operatorname{cosec}^2 y - \cot^2 y$

17. Evaluate $\sec^2 p - 1$.

A. $\tan p$

B. $\tan p \sec p$

C. $\tan^2 p$

D. $\tan^3 p$

18. Evaluate $\tan^2 q - \sec^2 q$.

A. 1

B. -1

C. 2

D. -2

19. What is the result of $\sec^2 x - \tan^2 x$?

A. 1

- B. -1
- C. 2
- D. -2

20. Which of these evaluates to -1 ?

- A. $\cot^2 r + \operatorname{cosec}^2 r$
- B. $\cot r$
- C. $\cot^2 r - \operatorname{cosec}^2 r$
- D. $\cot r - \operatorname{cosec} r$

21. Which of these is equivalent to $\sin 2w$?

- A. $2 \sin w \cos w$
- B. $2 \sin^2 w$
- C. $2 \cos^2 w$
- D. $\sin w \cos w$

22. Which of these is equivalent to $\cos 2z$?

- A. $1 - \cos^2 z$
- B. $1 + \cos z$
- C. $2 \cos^2 z - 1$
- D. $1 + 2 \cos^2 z$

23. Which of these is equivalent to $\tan 2t$?

- A. $\frac{1 - \tan^2 t}{t}$
- B. $\frac{2 \tan t}{1 - \tan^2 t}$
- C. $\frac{1 - \tan^2 t}{2 \tan t}$
- D. $\frac{\tan t + 1}{2 \tan t}$

24. Expand $\sin v$.

- A. $\sin \frac{1}{2} v \cos \frac{1}{2} v$
- B. $\sin^2 \frac{1}{2} v$

C. $\cos^2 \frac{1}{2}v$

D. $2\sin \frac{1}{2}v \cos \frac{1}{2}v$

25. Expand $\cos a$.

A. $\cos^2 \frac{a}{2} + 1$

B. $2\cos^2 \frac{a}{2} - 1$

C. $2\cos^2 \frac{a}{2} + 1$

D. $2 - \cos^2 \frac{a}{2}$

26. What is the equivalence of $\sin (p + q)$?

A. $\sin p \sin q + \cos p \cos q$

B. $\sin p \sin q - \cos p \cos q$

C. $\sin p \cos q - \cos p \sin q$

D. $\sin p \cos q + \cos p \sin q$

27. What is the equivalence of $\sin (p - q)$?

A. $\sin p \sin q + \cos p \cos q$

B. $\sin p \sin q - \cos p \cos q$

C. $\sin p \cos q - \cos p \sin q$

D. $\sin p \cos q + \cos p \sin q$

28. Expand $\cos (x + y)$.

A. $\cos x \cos y - \sin x \sin y$

B. $\cos x \cos y + \sin x \sin y$

C. $\sin x \sin y - \cos x \cos y$

D. $\sin x \sin y + \cos x \cos y$

29. Expand $\cos (x - y)$.

A. $\cos x \cos y - \sin x \sin y$

B. $\cos x \cos y + \sin x \sin y$

C. $\sin x \sin y - \cos x \cos y$

D. $\sin x \sin y + \cos x \cos y$

30. Express $\sin 60^\circ$ in surd form.

- A. $\frac{1}{2}$
- B. $\frac{\sqrt{3}}{2}$
- C. $\frac{1}{\sqrt{2}}$
- D. 0

31. Express $\sin 45^\circ$ in surd form.

- A. $\frac{1}{2}$
- B. $\frac{\sqrt{3}}{2}$
- C. $\frac{1}{\sqrt{2}}$
- D. 0

32. Which of these is equal to $\sin \theta$?

- A. $1 - \cos \theta$
- B. $\cos (90^\circ - \theta)$
- C. $\sin (90^\circ - \theta)$
- D. $\cos \theta - 1$

33. Which of these is equivalent to $\cos 13^\circ$?

- A. $\sin 77^\circ$
- B. $\cos 77^\circ$
- C. $\sin 87^\circ$
- D. $\cos 87^\circ$

34. Which of these is equivalent to $\cos \beta$?

- A. $\cos (90^\circ - \beta)$
- B. $\sin (90^\circ - \beta)$
- C. $\sec (90^\circ - \beta)$
- D. $\operatorname{cosec} (90^\circ - \beta)$

35. $\sin 81^\circ$ is the same as

- A. $\cos 9^\circ$
- B. $\sin 9^\circ$
- C. $\tan 9^\circ$
- D. $\sec 9^\circ$

36. What is the equivalence of $\tan (A + B)$?

- A. $\frac{\cos (A+B)}{\sin (A+B)}$
- B. $\frac{1}{\tan (A+B)}$
- C. $\frac{\tan A + \tan B}{1 - \tan A \tan B}$
- D. $\frac{\tan A - \tan B}{1 + \tan A \tan B}$

37. Which of the following is the same as $\tan (A - B)$?

- A. $\frac{\cos (A+B)}{\sin (A+B)}$
- B. $\frac{1}{\tan (A+B)}$
- C. $\frac{\tan A + \tan B}{1 - \tan A \tan B}$
- D. $\frac{\tan A - \tan B}{1 + \tan A \tan B}$

38. What is the equivalence of $\tan 90^\circ$?

- A. 0
- B. 1
- C. -1
- D. ∞

39. If $\cos \delta = \frac{a}{b}$, find $1 + \tan^2 \delta$.

- A. $\frac{b}{a}$

- B. $\frac{b^2}{a}$
 C. $\frac{a^2}{b}$
 D. $\frac{b^2}{a^2}+++$

40. If $\cos \delta = \frac{a}{b}$, find $1 - \tan^2 \delta$.

- A. $\frac{b}{a}$
 B. $\frac{2b^2 - a^2}{a}$
 C. $\frac{2a^2 - b^2}{a^2}+++$
 D. $\frac{a^2 - b^2}{b}$

41. If $\cos \delta = \frac{a}{b}$, find $\tan^2 \delta - 1$.

- A. $\frac{b}{a}$
 B. $\frac{b^2}{a^2}+++$
 C. $\frac{b^2}{a}$
 D. $\frac{a^2}{b}$

42. If $\cos \delta = \frac{a}{b}$, find $\operatorname{cosec}^2 \delta - 1$.

- A. $\frac{a^2}{b^2 - a^2}+++$
 B. $\frac{b^2}{b^2 - a^2}$
 C. $\frac{b^2}{a - b^2}$
 D. $\frac{a^2 - b}{b}$

43. If $\cos \delta = \frac{a}{b}$, find $\operatorname{cosec}^2 \delta + 1$.

- A. $\frac{2b^2 - a^2}{b^2 - a^2}+++$

B. $\frac{b^2}{b^2-a^2}$

C. $\frac{b^2}{a-b^2}$

D. $\frac{a^2-b}{b}$

44. If $\cos \delta = \frac{a}{b}$, find $\cot^2 \delta + \cos^2 \delta$.

A. $\frac{2b^2-a^2}{b^2-a^2}$

B. $\frac{a^4}{b^2(b^2-a^2)}$

C. $\frac{b^4}{a-b^2}$

D. $\frac{a^2-b}{b}$

45. Given that $\tan q = \frac{7}{2}$, evaluate $\sin 2q$.

A. $\frac{28}{35}$

B. $\frac{35}{28}$

C. $\frac{28}{53}$

D. $\frac{53}{28}$

46. Given that $\tan q = \frac{7}{2}$, evaluate $\cos 2q$.

A. $-\frac{53}{45}$

B. $\frac{53}{45}$

C. $\frac{45}{53}$

D. $-\frac{45}{53}$

47. Given that $\tan q = \frac{7}{2}$, evaluate $\tan 2q$.

A. $-\frac{45}{28}$

B. $\frac{54}{29}$

C. $\frac{29}{54}$

D. $-\frac{28}{45}$

48. Given that $\tan q = \frac{7}{2}$, evaluate $\sin 2q - \cos 2q$.

A. $-\frac{53}{73}$

B. $\frac{54}{29}$

C. $\frac{73}{53}$

D. $-\frac{28}{45}$

49. If $\sin \gamma = \frac{a}{1-b}$, find $\cot \gamma$.

A. $\frac{a}{(1-a-b)(1+a-b)}$

B. $\frac{(1-a-b)(1+a-b)}{a}$

C. $\frac{b}{(1-a+b)(1+a-b)}$

D. $\frac{(1-a+b)(1+a-b)}{b}$

50. If $\sin \gamma = \frac{a}{1-b}$, find $\cos \gamma$.

A. $\frac{(1-a-b)(1+a-b)}{1-b}$

B. $\frac{(1+a-b)(1+a-b)}{1-b}$

C. $\frac{(1-a-b)(1+a+b)}{1-b}$

D. $\frac{(1-a+b)(1+a-b)}{1-b}$